Daniel Hathcock

PhD Candidate in ACO · Carnegie Mellon University, Pittsburgh, PA

(+1) 303-834-2557 | dhathcoc@alumni.cmu.edu | https://danielhathcock.github.io

Education

Carnegie Mellon University — *Pittsburgh, PA*
Ph.D. in Algorithms, Combinatorics, and Optimization (ACO) — *Aug. 2020 - May 2026*

- Advised by Professor R. Ravi.
- Research in approximation algorithms for network design and resource allocation.
- NSF Graduate Research Fellowship recipient.

Carnegie Mellon University — *Pittsburgh, PA*
M.S. in Algorithms, Combinatorics, and Optimization (ACO) — *Aug. 2020 - Dec 2023*

- Awarded after completion of Ph.D. course requirements

Georgia Institute of Technology — *Atlanta, GA*
B.S. in Computer Science, Minor in Mathematics — *Aug. 2016 - May 2020*

- Concentrations in Theory and Intelligence.
- Undergraduate Thesis: Enumerating Acyclic Orientations, advised by Professor Prasad Tetali

Research Publications

Perfect Fractional Matchings in Bipartite Graphs Via Proportional Allocations — *preprint*
with R. Ravi — *2025*
CoRR preprint, 2025. arxiv.org/pdf/2510.01107

The Telephone k-Multicast Problem — *Algorithmica*
with Guy Kortsarz, R. Ravi — *2026*
Algorithmica, Volume 88, article number 25, (2026). doi.org/10.1007/s00453-026-01377-5

The Steiner Path Aggregation Problem — *IPL vol.192*
with Da Qi Chen, D Ellis Hershkowitz, R. Ravi — *2026*
Information Processing Letters, vol. 192:106608, 2026. arxiv.org/pdf/2510.01392

The Online Submodular Assignment Problem — *FOCS'24*
with Billy Jin, Kalen Patton, Sherry Sarkar, Mik Zlatin — *2024*
IEEE Symposium on Foundations of Computer Science (FOCS) 2024. arxiv.org/pdf/2401.06981

Approximation Algorithms for Steiner Connectivity Augmentation — *ESA'24*
with Mik Zlatin — *2024*
LIPIcs, Volume 308, European Symposium on Algorithms (ESA) 2024. arxiv.org/pdf/2308.08690
ESA best student paper award

The Telephone k-Multicast Problem — *APPROX'24*
with Guy Kortsarz, R. Ravi — *2024*
LIPIcs, Volume 317, APPROX/RANDOM 2024. arxiv.org/pdf/2410.01048

Maintaining Matroid Intersections Online — *SODA'24*
with Niv Buchbinder, Anupam Gupta, Anna R. Karlin, Sherry Sarkar — *2024*
ACM-SIAM Symposium on Discrete Algorithms (SODA24). arxiv.org/pdf/2309.10214

One Tree to Rule Them All: Poly-Logarithmic Universal Steiner Tree — *FOCS'23*
with Costas Busch, Da Qi Chen, Arnold Filtser, D Ellis Hershkowitz, Rajmohan Rajaraman — *2023*
IEEE Symposium on Foundations of Computer Science (FOCS) 2023. arxiv.org/pdf/2308.01199

Toppleable Permutations, Excedances and Acyclic Orientations — *CT-2(1)*
with Arvind Ayyer, Prasad Tetali — *2022*
Combinatorial Theory, 2(1). arxiv.org/pdf/2010.11236

On the Hypergraph Connectivity of Skeleta of Polytopes — *DCG'22*
with Josephine Yu — *2022*
Discrete & Computational Geometry (2022), pages 1–4. arxiv.org/pdf/2010.05053

Professional Experience

Teaching and Mentoring

Fall 2024	**Network Optimization I (47-835)**, Grader	*CMU*
Fall 2023	**Network Optimization I (47-835)**, Grader	*CMU*
Sum. 2023	**Polymath Jr. Research Program**, Graduate Mentor for Covering Grids with Hyperplanes Project	*Polymath Jr.*
Sum. 2022	**Polymath Jr. Research Program**, Graduate Mentor for Ramsey Theory Project	*Polymath Jr.*
Sum. 2021	**Concepts of Mathematics (21-127)**, Graduate Teaching Assistant	*CMU*
Spring 2021	**Discrete Mathematics (21-228)**, Graduate Teaching Assistant	*CMU*
Fall 2020	**Calculus in 3 Dimension (21-259)**, Graduate Teaching Assistant	*CMU*
Fa.'19 - Sp.'20	**Design & Analysis of Algorithms (CS 3510)**, Teaching Assistant	*Georgia Tech*
Fall 2018	**Intro to Linear Algebra (Math 1553)**, Teaching Assistant	*Georgia Tech*

Internships

Sum. 2020	**Tagup, Inc.**, Data Science Intern Developed and implemented machine learning models using JAX and Tensorflow for survival analysis, predicting time-to-event (TTE). Enabled >10x speedup of distributed model training using Dask.	*Somerville, MA*
Sum. 2017	**Left Hand Robotics**, Prototyping / Software Engineering Intern Worked on the design and control of a self-driving snow removal robot. Prototyped algorithms for high precision GPS path collection, following, and correction. Used Python and Java being run on a Raspberry Pi.	*Longmont, CO*

Professional Service

2023	**CMU Theory Lunch Seminar**, Co-organizer	*CMU*
Fa.'19 - Sp.'20	**Georgia Tech Theory CS Club**, Talk Coordinator	*Georgia Tech*

Honors & Awards

Academic

2024	**ESA best student paper award**, for 'Approx. Algorithms for Steiner Connectivity Augmentation'	*London, UK*
2021	**NSF Graduate Research Fellowship Program**, Awarded (3 years PhD funding)	*NSF*
2020	**Highest Honor**, Georgia Tech institutional honors	*Atlanta, GA*
2016-2020	**Faculty Honors Letter**, 4.0 GPA all semesters at Georgia Tech	*Atlanta, GA*

Professional

2022	**CMU ACM Hackathon: Algorithms with a Purpose**, Second Place	*Pittsburgh, PA*
2021	**CMU ACM Hackathon: Algorithms with a Purpose**, Second Place	*Pittsburgh, PA*
2018	**HackGT: Goldman Sachs Data Mining Challenge**, First Place	*Atlanta, GA*
2017	**HackGT: FINRA Data Mining Challenge**, First Place	*Atlanta, GA*

Misc

Languages	Python/SageMath, Mathematica, Java, C/C++, LaTeX
Skills/Tools	Scientific Python (SciPy, NumPy, Scikit, Jupyter, etc.), Gurobi, OR-Tools, Git, GitHub Copilot, TensorFlow, PyTorch, Dask
Service	Math Tutoring at Minadeo Elementary, a Title I school in Pittsburgh, PA (2022)